

DASC7606C Group Project

Last edited: 24 September 2026

Overview	1
GPU Resources	2
Policy on LLM Usage	2
Topics	3
Track 1: AI Startup Business Proposal 🌟	4
Requirements	4
Deliverables	4
Scoring	4
Track 2: Agentic Framework Design 🌟🌟	5
Requirements	5
Deliverables	6
Scoring	6
Track 3: Simulation Experiments 🌟🌟	7
Resources	7
Requirements	7
Suggested Research Questions	8
Deliverables	8
Scoring	9
Track 4: Real-robot Training & Deployment 🌟🌟🌟	9
Further Clarification about Grading	9

Overview

The group project accounts for **25%** of your final grade in DASC7606C. Based on initial enrollment, the class will be divided into **10 groups** of approximately **10–12 students**. Group sizes may be adjusted if enrollment changes.

We recommend forming your group after the first class. Once you have agreed on your group members and a tentative topic, nominate one representative to register through the Google Form [DASC7606C Group Project Registration](#). Submit one registration per group. If your group is not yet full, list your current members and indicate how many additional members you need. The teaching team will help coordinate group allocation. Or you can also join others' groups you are interested in this way.

After the teaching team confirms your group registration, submit one proposal per group covering:

- Your chosen project track, specific topic, and objectives.
- Each member's role and responsibilities.
- Your planned approach and expected deliverables.

Before Lecture 5.

All group members must review and agree to the proposal before submission. The proposal will be reviewed for scope and feasibility but will not receive a separate grade. The TAs will contact your group if revisions are needed.

Your topic is fixed once the proposal is approved. Any subsequent topic change requires approval from the teaching team. You may refine your methods and deliverables as the project develops, provided they remain consistent with the approved topic.

Please follow the registration and proposal **deadlines** announced on **Moodle**.

If you have any questions, please contact:

- All Staff: dasc7606c-staff-2026fall@googlegroups.com
- Head TA: teaching@bangjunwang.com

GPU Resources

- Option 1: You can apply for access to the HKU CS GPU Farm (GPU Farm for Teaching, formerly Phase 2) - RTX 4080 Cluster. Please refer to <https://www.cs.hku.hk/research/major-facilities?view=article&id=130&catid=24> for resource applications and usage.
- Option 2: DASC7606C is supported by [Moore Threads](#) by offering our students with higher standard state-produced GPU resources. **If you can integrate Moore Threads GPUs into your group project and include the profiling and benchmarking results in your final report, your group will get a 5-point bonus directly on the final project group score.**

Update: Lecture 5 will cover GPU fundamentals and include a hands-on tutorial on MUSA programming. Please make sure to attend.

If you have any questions regarding GPU resources, please contact Serena for help via serena.cds@hku.hk

Policy on LLM Usage

You may use large language models (LLMs) for project ideation, writing, coding, and analysis. In fact, we encourage you to embrace AI tools while using them thoughtfully and checking their outputs carefully.

Adapted from the [ICLR 2027](#) policy, the following rules apply:

1. Disclose all LLM use. Include an “LLM Usage Statement” in your proposal and final report. Identify the tools used, the specific sections or components affected, and how the tools contributed. This includes grammar and wording improvements. If no LLMs were used, state this.
2. Take responsibility for your work. Your group is responsible for all submitted content and must verify LLM-generated claims, references, code, and results. Using an LLM does not excuse false claims, plagiarism, fabricated data, or misrepresented results.
3. Protect confidentiality. Do not upload confidential information or restricted data to an LLM without authorization.
4. Do not manipulate assessment. Hidden prompts or instructions intended to influence grading are prohibited, whether or not they succeed.

Disclosed LLM use will not itself reduce your grade. Violations will be handled under the course and institution’s academic integrity policies.

We will conduct strict checks for AI-related plagiarism and undisclosed AI use. Suspected violations will be reviewed using supporting evidence, rather than AI-detection scores alone. Confirmed misconduct may result in substantial penalties to your final course grade, including a zero for the group project, and further disciplinary action under the University’s academic integrity policies.

Example LLM Usage Statement

- Section 2 — Market Analysis: We used ChatGPT to identify potential competitors. We checked the product descriptions and pricing included in the report against official sources.
- Section 3 — Technical Approach: We used Claude to compare candidate model architectures. The group selected the final approach and verified the cited references.
- Prototype — preprocess.py: We used ChatGPT to draft the data-cleaning function. We modified the function and tested its handling of missing values and duplicate records.
- Sections 1 and 5 — Writing: We used ChatGPT to improve grammar and clarity. The group reviewed and edited the suggested wording.

Your statement should describe your group's actual use; the examples above are illustrative.

Topics

This group project offers four tracks. The number of 🌟 indicates each track's difficulty and potential for higher marks. Choosing a more difficult track does not guarantee a higher score: an excellent submission in a simpler track can outperform one that only meets the basic requirements of a harder track. Your grade will depend on the quality of your work and how well it meets the assessment criteria.

Track 1: AI Startup Business Proposal 🌟

Propose an AI startup whose core product uses deep learning. Explain the customer problem, how the technology solves it, and how the startup could earn revenue.

Requirements

Your proposal must cover:

1. Problem and customers: Who needs the product? What problem does it solve? Provide evidence that the problem matters.
2. Product: Describe the product, how customers would use it, and its advantages over existing alternatives.
3. Deep learning approach: Explain the model, inputs and outputs, data sources, and evaluation metrics. Justify your technical choices.
4. Business model: Explain pricing, customer acquisition, and estimated development and operating costs. State your assumptions.
5. Feasibility and risks: Outline an MVP, development milestones, and major technical, business, or ethical risks.

Pretrained models and commercial APIs are allowed. A working prototype or demo is encouraged but not required. However, the proposal must include evidence of technical feasibility, such as results from a small experiment or relevant published research.

Deliverables

- Report: 8 -10 pages, excluding references.
- Pitch: 10-minute presentation followed by 5-minute Q&A.
- Contribution statement: A brief description of each member's work.
- LLM disclosure.

Scoring

Criterion	Points	What we assess
Problem and customer need	20	Clear target customers and evidence of a meaningful problem
Product and competitive advantage	15	Useful product and convincing comparison with alternatives
Deep learning approach	25	Appropriate models, credible data plan, and sound evaluation
Business model	20	Plausible pricing, customer acquisition, and costs
Feasibility and risks	10	Realistic scope, supporting evidence, and awareness of risks
Report, presentation, and Q&A	10	Clear explanation and well-supported answers
Total	100	

Cite external claims and clearly label assumptions. Marks reward evidence and sound reasoning; ambitious claims and polished slides alone will not earn high scores.

Track 2: Agentic Framework Design 🌟🌟

Design and build an AI tool that solves a specific problem. Explain who it serves, how it works, and demonstrate its usefulness through a working demo.

Requirements

Your project must cover:

1. Problem and users: Who will use the tool? What problem does it solve? Explain why the problem matters.
2. Tool design: Describe the main features, user workflow, and advantages over existing alternatives.
3. Deep learning approach: Explain the models, inputs and outputs, data sources, and evaluation metrics. Justify your technical choices.
4. Implementation and evaluation: Build a working tool and evaluate it using representative examples or test cases. Report what works and where it fails.
5. Limitations and risks: Discuss technical limitations, privacy concerns, and other relevant risks.

Pretrained models and commercial APIs are allowed. Your contribution should go beyond a basic model or API call by providing a usable workflow for the chosen problem.

A working demo is required and must be shown during the presentation. Demonstrate the complete workflow, from user input to the resulting output. Prepare a backup recording in case of technical problems.

Deliverables

- Report: 5–8 pages, excluding references.
- Presentation: 10 minutes, including the demo, followed by 5 minutes of Q&A.
- Code: Source code with setup instructions and any necessary sample inputs. Do not include API keys or restricted data.
- Contribution statement: A brief description of each member's work.

Scoring

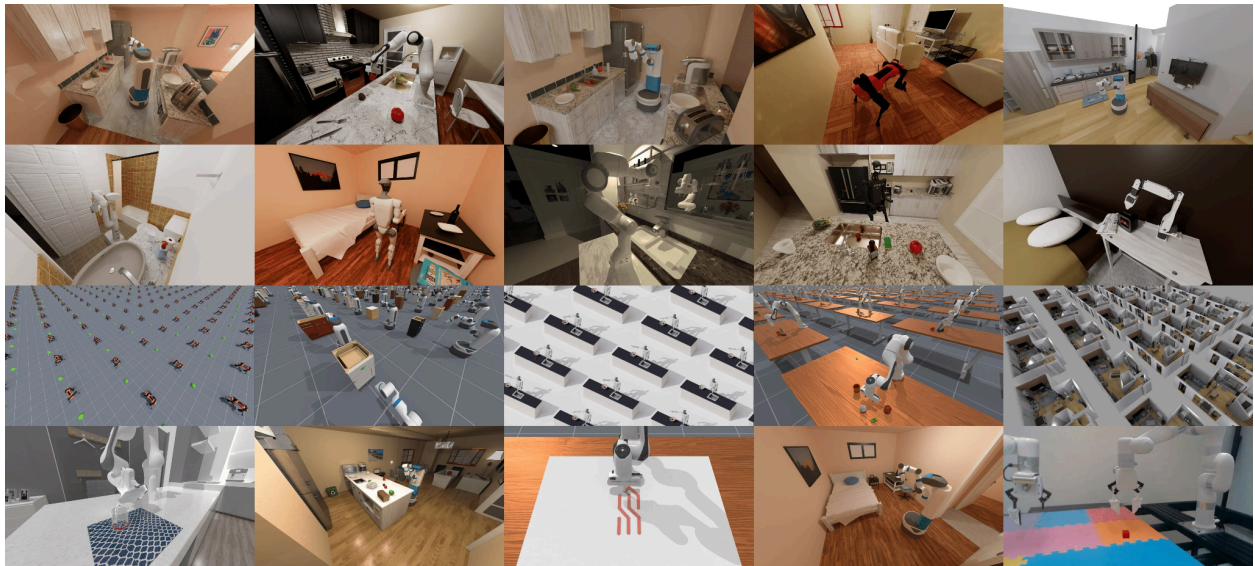
Criterion	Points	What we assess
Problem and user need	15	Clear users and a meaningful problem
Tool design and usability	15	Useful features and a coherent user workflow
Deep learning approach	20	Appropriate models, justified choices, and a credible data plan
Implementation and demo	25	A functioning tool that demonstrates the intended workflow

Evaluation and limitations	15	Meaningful testing, evidence of usefulness, and honest analysis of failures and risks
Report, presentation, and Q&A	10	Clear explanation and well-supported answers
Total	100	

Cite external sources and follow the course's LLM usage policy. Marks reward a useful, reliable tool and sound technical reasoning. A complex system does not automatically earn a higher score.

Track 3: Simulation Experiments 🌟🌟

Collect or generate expert demonstrations and train and evaluate **Diffusion Policies (DP)** on at **least six distinct ManiSkill tasks**. Investigate a research question beyond basic training and demonstrate your findings.



Resources

- ManiSkill: [Official repository](#) · [Documentation](#)
- Diffusion Policy: [Original repository](#) · [Project website](#) · [Paper](#)

Use the documentation matching your installed ManiSkill version. The original DP repository is a reference implementation; integration with ManiSkill may require adaptation.

Requirements

1. Task selection: Select at least six distinct tasks covering different manipulation skills. Different random seeds of the same task do not count as separate tasks. Describe the observations, action spaces, and success criteria.
2. Expert demonstrations: Collect or generate demonstrations for all six tasks using teleoperation, motion planning, scripted controllers, or another justified method. Document the process, dataset sizes, and quality checks. Existing tools are allowed, but downloading datasets alone does not satisfy this requirement.
3. Policy training: Train and evaluate a DP baseline for every task. Separate task-specific policies are acceptable. Document the architecture, preprocessing, training settings, and computing resources.
4. Further investigation: Explore at least one question from the suggestions below, or propose your own. This investigation may focus on a justified subset of tasks.
5. Evaluation and analysis: Use held-out evaluation seeds or initial conditions. Report per-task success rates, evaluation episode counts, and failure cases. Keep comparisons fair and state any differences in data or compute budgets.

Suggested Research Questions

- Co-training: Does training one policy on multiple tasks help or hurt performance compared with separate policies? How do task similarity, data balance, and task conditioning affect the results?
- Improving DP: Can changes to the observation encoder, prediction horizon, action representation, or diffusion sampling improve success rates or inference speed? Test individual changes against the baseline.
- Cross-embodiment: Can demonstrations from one robot help train a policy for another? Explore how to handle different observations, action spaces, and robot geometries. Scope this study to compatible tasks.
- Data efficiency: How many demonstrations are needed? Does demonstration diversity or quality matter more than quantity?
- Generalization: How well does the policy handle changes in object positions, appearances, or physical properties? Can augmentation or domain randomization improve robustness?

You are not expected to explore every direction. A focused investigation with clear evidence is sufficient, even if the proposed change does not improve performance.

Deliverables

- **Report:** 5–8 pages, excluding references.

- **Presentation:** 10 minutes, including a **required demo**, followed by 5 minutes of Q&A. Show the demonstration-generation process and policy rollouts, with video coverage of all six tasks. Recordings are acceptable.
- **Code, data, and policies:** Collection scripts, generated datasets or download links, training and evaluation code, configurations, and trained checkpoints, with reproduction instructions.
- **Contribution statement:** A brief description of each member's work.

Scoring

Criterion	Points	What we assess
Task coverage and demonstration generation	20	Six distinct tasks, a working collection process, and documented data quality
DP implementation and baseline training	20	Correct implementation and documented training across all six tasks
Further investigation	20	A clear question, justified approach, and controlled experiments
Evaluation and analysis	25	Fair comparisons, per-task results, and thoughtful interpretation of failures
Demo and reproducibility	5	Clear demonstrations and usable artifacts
Report, presentation, and Q&A	10	Clear explanation and well-supported answers
Total	100	

Follow the course's LLM usage policy and acknowledge external code and data. Marks reward experimental rigor and understanding; high success rates alone do not guarantee a high score.

[UPDATED] Track4: Humanoid-robot Motion Generation

Deployment 🌟🌟🌟

Explore how natural-language instructions can control a Unitree G1 humanoid robot. Each group will experience real-robot data collection and inference during a supervised visit to Archon, then build and evaluate a text-to-motion system in simulation. The simulation project must use SONIC as its low-level controller and demonstrate tabletop block grasping.

Resources

- Motion-generation references: [Kimodo](#) · [ARDY](#) · [MotionBricks](#) · [Awesome Text-to-Motion collection](#)
- GEAR-SONIC: [Official repository](#) · [Documentation](#)
- Unitree G1: [Robot description](#) · [SDK2](#)
- LeRobot: [Official repository](#) · [Dataset documentation](#)

These repositories are references rather than required model choices. Groups should assess what adaptation is needed to execute generated motions through SONIC and interact with objects.

Requirements

1. Archon session: Book a supervised visit to Archon at least 2 days in advance. During the session, use the provided system to go through the real humanoid robot's demonstration-recording and policy-inference workflows. Explain the inputs, recorded data, model outputs, and robot control flow in your report. This session does not require groups to train a new policy or deploy their own text-to-motion system on the robot.
2. SONIC integration: SONIC must be the low-level motion controller for the group's text-to-motion system. Explain how generated motions are represented, adapted to the G1, and executed through SONIC. Document any separate hand or gripper control.
3. Text-driven grasping: In simulation, use natural-language instructions to make the G1 grasp and lift a block from a table. Step-by-step commands for reaching, adjusting the arm or hand, grasping, and lifting are acceptable. Typed text is sufficient; speech input may be added through speech-to-text. Existing simulation scenes may be reused.
4. Method comparison: Implement and compare at least two distinct motion-generation methods using the same SONIC controller. Describe each method's inputs, outputs, integration, and any additional planning or correction components. Changing only the prompt for one method does not count as a second method.
5. Evaluation and analysis: Test both methods with a shared set of instructions and comparable simulation conditions. Define success criteria and report repeated-trial grasp success, response time, the number of user corrections, and representative failures. Analyze instruction following, motion precision, physical feasibility, and object interaction. Explain how the lower body is stabilized if only the upper body is controlled.

Suggested Research Questions

- How do the methods differ in their ability to follow text instructions and reach a specified grasp pose?
- Do spatial constraints or interactive corrections improve grasp success?
- What trade-offs exist between motion quality, response time, and ease of integration with SONIC?

A focused comparison with clear evidence is sufficient, even if neither method achieves a high grasp success rate.

Optional Extensions

- **Sim-to-real text-to-motion:** Submit simulation videos, evaluation results, and a deployment plan to the TAs sufficiently in advance for review and revision. After approval, book a supervised session at Archon to test the text-to-motion system on the physical robot. Robot access is subject to availability and laboratory safety procedures.
- **Simulation data collection:** Build a tabletop scene with a table and graspable objects. Use text-driven control to collect synchronized robot actions, robot states, and images from a simulated head-mounted camera. Groups may choose and document the camera pose and intermediate file format, but must convert the final episodes to LeRobot format with instructions, timestamps, and episode boundaries. Upper-body data is sufficient.
- **Generalization or model improvement:** Replace object assets, vary object positions or orientations, or fine-tune a motion-generation model to improve performance. Compare results against the original setup using repeated trials.
- **New tasks or whole-body data:** Build another simulated task or extend text-to-motion control to the whole body.

The extensions above are examples, not an exhaustive list. Groups may pursue other original ideas aligned with this track. Extra credit will be awarded according to the completeness, technical quality, and supporting evidence of the finished work.

Deliverables

- **Report:** 5–8 pages, excluding references. Describe the Archon session, simulation system, compared methods, evaluation, failures, and any extensions.
- **Presentation:** 10 minutes with slides and a required demonstration, followed by 5 minutes of Q&A. Recorded demonstrations are acceptable.
- **Videos:** Show the text instructions and corresponding robot behavior, including successful and failed trials for both methods. Include evidence of any completed extensions.
- **Public GitHub repository:** Include its link in the report and presentation. Publish the group's implementation, configurations, setup instructions, and evaluation code. For data-collection extensions, provide recording and LeRobot conversion scripts and the dataset or an accessible link. Do not publish credentials, restricted robot data, or third-party materials without permission.

Scoring

Criterion	Points	What we assess
Archon session and workflow understanding	10	Participation in the supervised session and accurate explanation of data collection and inference

SONIC integration	15	Correct use of SONIC and a clear motion-to-control pipeline
Text-driven grasping	20	A working simulated grasp-and-lift workflow controlled by text
Motion-generation comparison	20	Two distinct methods, justified choices, and fair experimental conditions
Evaluation and analysis	15	Repeated trials, clear metrics, failure cases, and sound interpretation
Demo and reproducibility	5	Clear videos and a usable public GitHub repository
Report, presentation, and Q&A	15	Clear explanation and well-supported answers
Total	100	

Follow the course's LLM usage policy and acknowledge all external code, models, and datasets. Marks reward a complete and safe real-robot learning pipeline, experimental rigor, and technical understanding; a polished demonstration or high success rate alone does not guarantee a high score.

Further Clarification about Grading

The group project will be assessed by the instructor and TA panel, who will assign an overall score to each group. Each member's individual score will be calculated by multiplying the group score by a contribution factor, based on their contribution statement and supporting evidence. **Scores will not be announced publicly.** Once officially communicated, individual scores will be considered final, subject to the University's applicable review procedures. We encourage everyone to contribute actively, keep a clear record of their work, and raise concerns about workload distribution early.